

W4

BLOCK 1 · CLASS 4 · PRACTICAL

Rules, judge, loop.

Last week you watched a feature get built. Today you build one.

Today.

- 1 Check-in
- 2 Git: one change, start to finish
- 3 From a model to an agent
- 4 Three standards
- 5 Build one, live — then you build one
- 6 Retro: one new rule

Your GitHub page.

- A branch for feature 1.
- A merged pull request.

Not there yet? Start there today.



HOW A TEAM STAYS SANE

One change, start to finish.

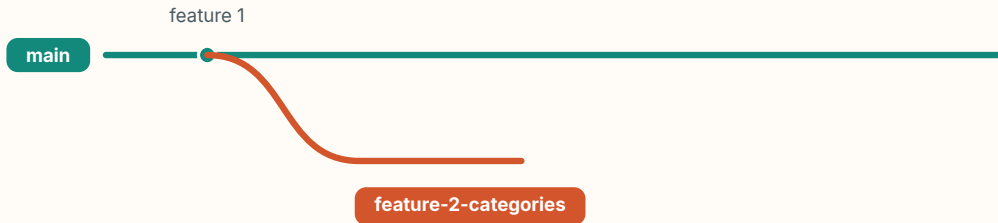
`main` is the version everyone trusts. Here is how a change gets into it.

main



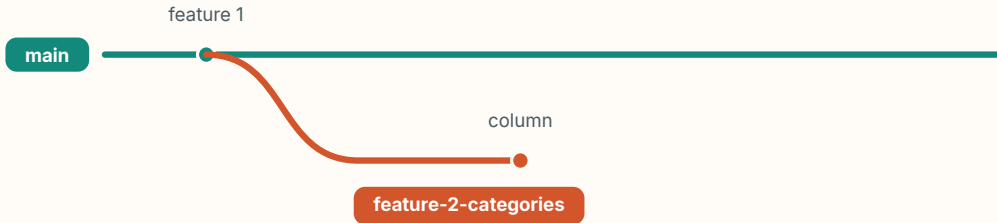
One version, one history, one truth. No folder full of *model_v3_FINAL_jm*.

A branch.



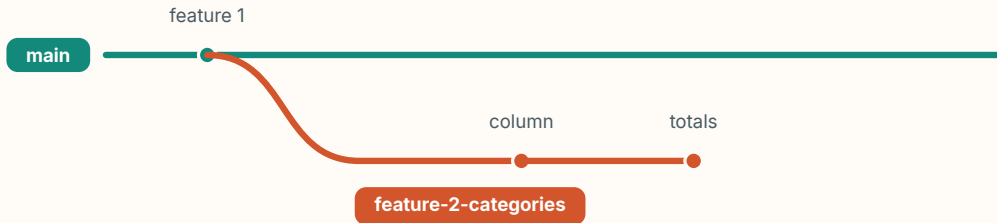
A **branch** is your own copy to work in, named after the job. Everyone else carries on from `main`, undisturbed.

A commit.



A **commit** is a checkpoint with a note saying what changed. Small ones. You can go back to any of them.

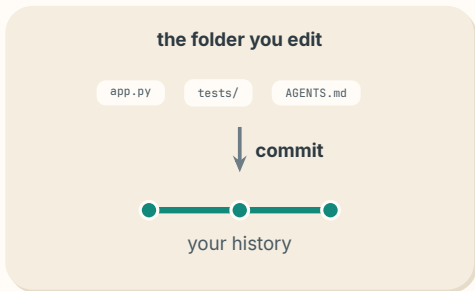
Working on a branch.



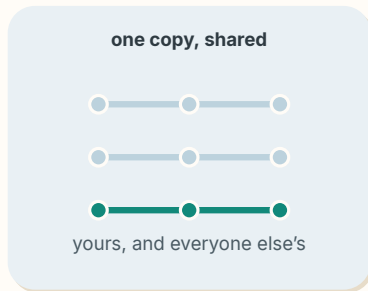
Three commits in and half of it broken. Everyone else's copy is untouched.

Local, and remote.

YOUR LAPTOP · LOCAL



GITHUB · THE REMOTE

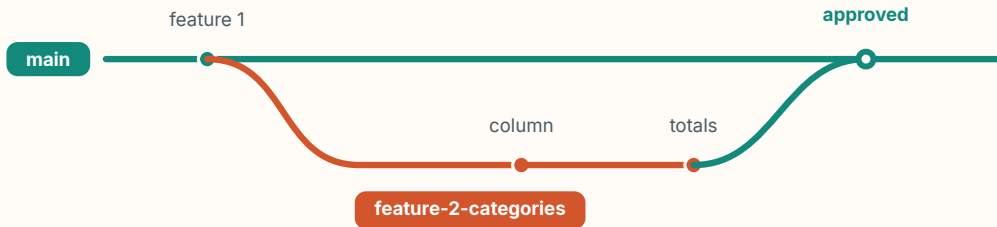


push
→
your branch, up

pull
←
their work, down

Push sends your branch to the copy everyone shares, where a **pull request** asks someone to read the diff first. Today that someone is you, reviewing the agent.

Merge.



Merge is the sign-off. **main** now has your work, and the next person starts from there.

agent

FROM A CHAT MODEL TO AN AGENT

What is actually running.

A model that can only write text — and the software around it that lets it act.

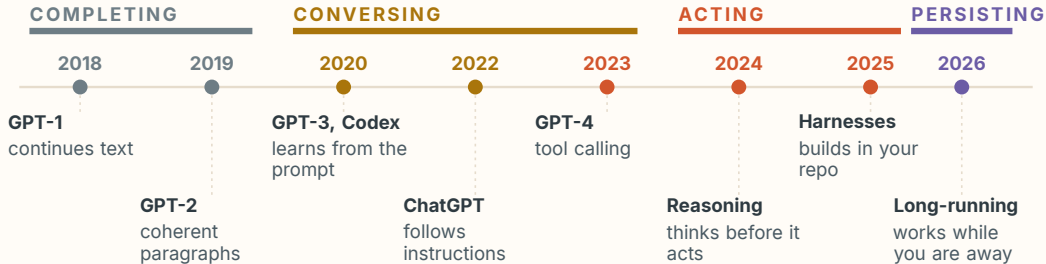
The long run-up.



The ideas were all there. What was missing arrived after 2007: **cheap parallel arithmetic** (graphics cards) and **the web as text**.

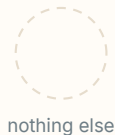
NVIDIA CUDA (2007) · Krizhevsky, Sutskever & Hinton, AlexNet (2012) · Common Crawl (public crawls from 2011)

2018 onwards.



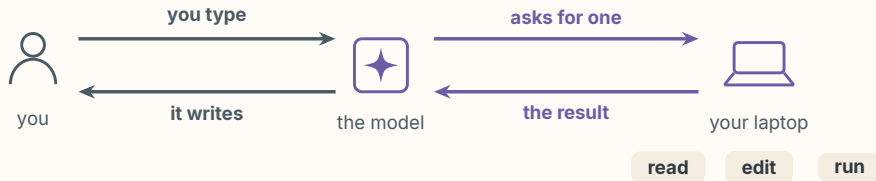
The job changed twice: when it learned to take **instructions**, and when it got **tools and a loop**. The next four slides are that second change.

A language model.



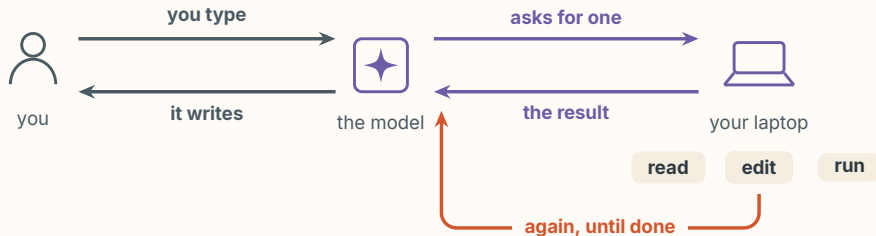
Text in, text out. It has never seen your folder and cannot run anything. In Class 2, **you** were the hands.

Tools.



A **tool** is a named action it may ask for. Three cover nearly everything. The result comes back to **it**, not to **you**.

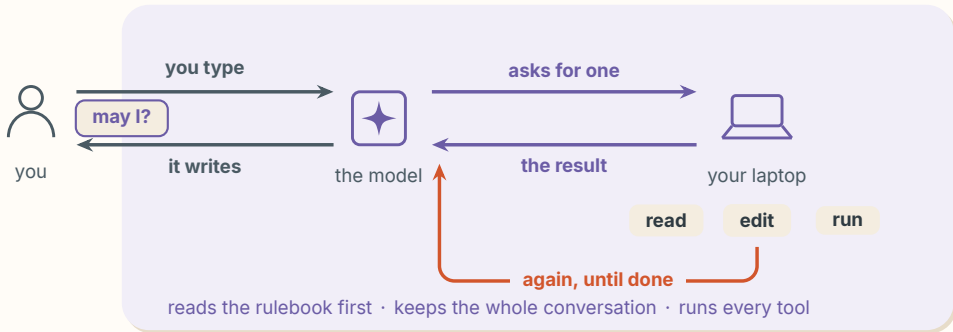
An agent.



Agent = model + tools + a loop. It acts, sees the result, and chooses again.

The harness.

CODEX · THE HARNESS



The software around the model. Codex is one; Claude Code and Cursor are others. Same loop everywhere — **learn it once**.

One exchange, in full.

YOU TYPE what was the last expense?

THE MODEL EMITS `{"tool": "read_file", "path": "data/expenses.csv"}`

CODEX ASKS *Codex wants to read* `data/expenses.csv` *. Allow?*

CODEX PASTES BACK `29,3,8,Jantar no Zé dos Cornos,30.00,2026-08-25`

THE MODEL ANSWERS €30 — dinner at Zé dos Cornos, on 25 August.

It never saw the file. It asked, you said yes, and the line came back.

Reasoning.

YOU

Make it so Ana doesn't owe anything.

THINKING

That could mean delete her expenses, mark her settled up, or take her out of the split. Two of those change what everyone else owes. One doesn't.

CODEx

Marked Ana settled up, so the other balances are untouched. Deleting her expenses would move theirs too — tell me if you meant that.

The loop has a name: ReAct.



Reason, act, observe, repeat. Every harness since is this loop with better plumbing.

Yao, Zhao, Yu, Du, Shafraan, Narasimhan & Cao (2022). *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR 2023 · arxiv.org/abs/2210.03629

WHAT IT CAN HOLD

Context.

EVERYTHING IT CAN SEE AT ONCE



WHAT IT IS

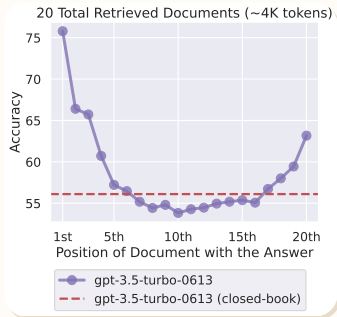
The whole conversation: every file it opened and every command it ran is still in there, taking up room.

WHY IT MATTERS

Fixed size. When it fills, the oldest part goes first — usually your instructions.

Context rot.

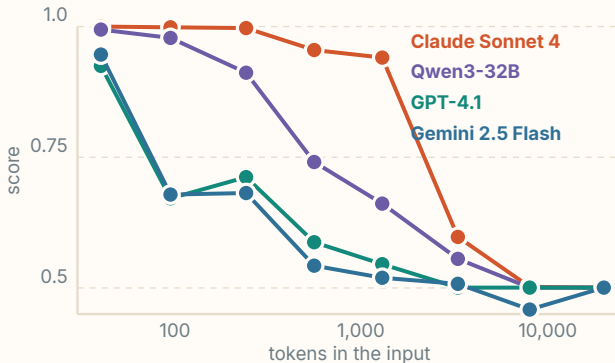
WHERE THE ANSWER SITS



Answer in the middle: **worse than the dashed line**, which is no documents at all.

Liu et al. (2024), *Lost in the Middle*, TACL 12, fig. 1, as published (CC BY 4.0) · right redrawn from Chroma (2025), *Context Rot* (MIT); 18 models in full

HOW LONG THE INPUT IS



Four 2025 models copying a list of words. Every one falls from a few hundred tokens on — nowhere near the window.

3

THE CORE OF THE CLASS

Three standards.

A rulebook it reads. A judge it cannot argue with. A loop that knows when to stop.

AGENTS.md

WHAT IT IS

A README for agents.

Every session starts by reading it.

WHAT GOES IN

What it cannot guess.

Commands. Boundaries.
How to report back.

WHAT STAYS OUT

What it can read.

Your code, explained.
Every extra line hides a real one.

It lives in git. When the agent misbehaves, the fix is usually a line here.

Three rules.

BOUNDARY

One branch per spec.

Never straight to `main`.

EVIDENCE

Show me the test output.

The output itself, not a summary of it.

OFF-LIMITS

Never touch the tests.

If one looks wrong, say so and stop.

Exact wording on the Week 4 page. Add them, read the diff, commit, push.

The judge.

THE PROBLEM

So give it a check.

Failing tests are the to-do list. Green is the finish line.

THE RULE

Editing a test is cheating.

It is not finishing.

EVIDENCE

Show me the output.

Anyone can type "all tests pass".

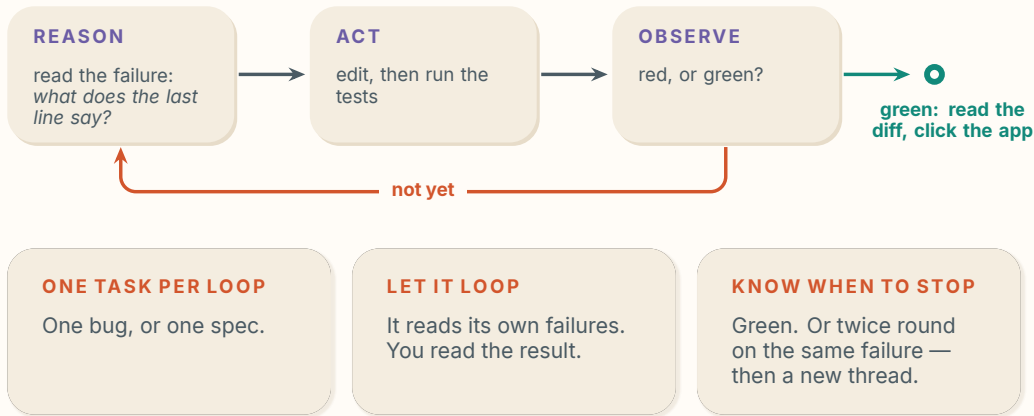
AFTER GREEN

Read the diff. Click the app.

App wrong, tests green? You found a missing test.

Decide what *done* means before you start.

The loop.



The shape of an ask.

1

The task

One thing.
Name the spec.

2

The boundary

This branch.
Never the tests.

3

The judge

Run the tests
until they pass.

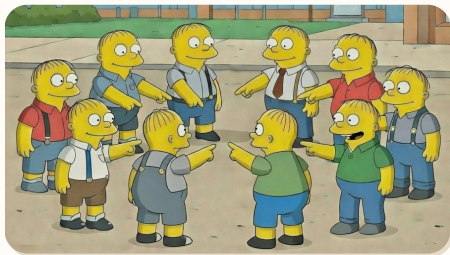
4

The evidence

The output, and
every file
touched.

Written out in full on the Week 4 page. Copy it; change two names.

Ralph.



WHAT IT IS

The same prompt, handed to a fresh agent again and again, until the judge says green.

WHY A CROWD

Every run is a new one, and none remembers the last. What they need to know has to be written down.

Nobody is watching.

```
while ;; do cat PROMPT.md | claude-code ; done
```

Published in 2025. That is the entire technique.

THE PROMPT

One file, read again from scratch every time. It is the only instruction anyone gives.

THE REPO

Nothing survives a run except what was written to disk. The rulebook and the tests are the memory.

THE CATCH

Nobody reads anything until it stops. A weak judge buys you hours of confident wrong.

/goal

```
/goal every test in tests/ passes, and nothing in tests/ changed
```

2026. The shell loop is now one sentence, in the tool.

WHO DECIDES

After each turn a *second*, smaller model reads the conversation and rules: not yet, done, or impossible. Not the one doing the work.

WHAT IT CAN SEE

It cannot run your tests. It only reads what the agent put on screen. So the finish line has to be something the agent can show you.

STILL YOURS

The tool runs the loop now. Nobody has automated deciding what *done* means.

Feature 2, live.

- 1 **Branch.** Named after the spec.
- 2 **Red first.** See what is missing before anything is written.
- 3 **New thread. Paste the ask.** Then sit on your hands.
- 4 **Watch it loop.**
- 5 **Green — now don't trust it.** Read the diff. Click the app.
- 6 **Commit, push, pull request, merge.**

One bug, then feature 2.

SPLITIT

Bug 2, then feature 2

Removing a member crashes the page.
Then expense categories.

TINY CRM

Bug 3, then feature 2

"Won this month" is always zero. Then
conversion by source.

Files, test names and the ask: Week 4 page.

When it goes wrong.

- 1 **It edited a test.** Discard. Say the rule. Run again.
- 2 **"Done" — tests red.** Ask for the output.
- 3 **Same failure, three times.** Stop. Read it yourself. New thread.
- 4 **It fixed three other things.** Discard. One task per loop.
- 5 **Green, app wrong.** Write down what the judge missed.

Every line here is a candidate rule for your `AGENTS.md`.

Write one rule.

- The worst thing it did today — one sentence.
- Turn it into a rule you could check.
- Commit it to `AGENTS.md`, with the reason.

A rulebook grows one line at a time, from things that went wrong.

Before next week.

- Feature 2: green, merged, pushed.
- One new rule in `AGENTS.md`.
- Push before you close the laptop.

Rulebook · judge · loop.